

Heena Khan

Phone: (516) 613-1854 Email: henakhan@stanford.edu GitHub: [heenakhan122](https://github.com/heenakhan122) Portfolio: heenakhan122.github.io/Heena-Khan-Portfolio/

EDUCATION

Stanford University

Expected Graduation: June 2027 (B.S.)

B.S. Computer Science (Artificial Intelligence) | Minor: Biology

Relevant Coursework: Operating Systems, Compilers, Data Structures & Algorithms, Databases, Probability & Statistics, Machine Learning, Discrete Mathematics, Cloud Computing, Computer Vision

EXPERIENCE

Software Engineering Intern — Data Infrastructure & Full Stack

Santa Clara, CA | June 2026 – October 2026

Roche / Genentech

- Deployed NL-to-SQL AI agent (LLM-backed) with schema-grounded prompting over 25+ tables and a ground-truth Q&A evaluation framework scoring SQL syntax validity and execution correctness; paired with a custom React chat UI.
- Designed 25-table PostgreSQL schema for enterprise PCR/HiT-PCR assay data using class-table inheritance (`result` → `pcr_result` → `hit_result`), design-time/run-time segregation, and junction tables for pool membership.
- Engineered Django Ninja REST API with typed Pydantic schemas and React + Vite dashboard featuring real-time PCR amplification curve visualizations (Recharts) and ETL audit log viewer.
- Built 5 Python ETL ingestion pipelines processing multi-sheet Excel reports and instrument CSVs into PostgreSQL with full audit logging via `etl_run_log`, deployed to Linux sHPC virtual machine.

Haas Center for Public Service Fellow

Stanford, CA | June 2023 – September 2023

Afghanistan Women's Council

- Built full-stack e-commerce platform end-to-end — React + TypeScript frontend, Node.js/Express REST API, PostgreSQL with Drizzle ORM — enabling 100+ Afghan women artisans to sell handmade goods globally.
- Integrated Stripe payments with fraud controls, Zod server-side validation, and automated CI/CD via GitHub Actions; sole engineer from architecture through production deployment.

Stanford Tech Fellowship

Stanford, CA | 2024

Pull for Progress

- Engineered geospatial ML models in Python using multidimensional health metrics to predict regional risk profiles across 50,000+ healthcare data points.
- Applied statistical modeling and spatial feature extraction to optimize healthcare worker deployment in resource-limited settings.

PROJECTS

AMANI — AI-Enabled Health Mapping System | Stanford Healthcare Design Challenge Finalist

React Native, TensorFlow Lite, PostgreSQL, REST APIs, Computer Vision

- Led development of offline-first mobile app integrating on-device computer vision to classify WASH hazards in low-connectivity environments; quantized TFLite models to run on low-RAM Android devices.
- Designed privacy-preserving system with anonymous reporter IDs, AES-256 encrypted sync payloads, and human-in-the-loop verification — AI suggests classification, frontline staff confirm before dispatch.
- Built offline sync queue with exponential backoff and geo-proximity deduplication to preserve data integrity across intermittent connectivity.

Khaista Boutique — Full-Stack E-Commerce Platform

TypeScript, React, Node.js, PostgreSQL, Drizzle ORM, Stripe, Vite

- Built end-to-end e-commerce platform with React + TypeScript frontend, Node.js/Express REST API, and PostgreSQL with Drizzle ORM for Afghan women artisans selling handmade goods globally.
- Integrated Stripe payments, server-side input validation with Zod, and automated CI/CD via GitHub Actions with environment-specific deployments.

ModestFilter — Chrome Extension (Manifest V3)

TypeScript, React, Chrome APIs, MutationObserver, chrome.storage

- Built Manifest V3 Chrome extension in TypeScript and React with real-time keyword filtering across 7+ e-commerce sites, MutationObserver-based DOM handling for infinite scroll, and chrome.storage sync for persistent preferences.

SKILLS

Languages: Python, SQL (PostgreSQL), TypeScript, JavaScript, C++, Java

Infrastructure & Backend: Django / Django Ninja, Pydantic, REST APIs, PostgreSQL (schema design, inheritance, indexing), ETL Pipelines, Docker, Linux/Bash, CI/CD

Frontend & ML: React, Vite, Recharts, Pandas, NumPy, Scikit-Learn, TensorFlow Lite, LLMs / NL-to-SQL